

USHNA - Project Progress Report

Date: 2026-09-16

1. Project Initialization & Structure

The USHNA project repository has been initialized with a professional Git setup. A modular directory structure was created to separate concerns: 'src/physics', 'src/learning', 'src/optimization', 'src/data', and 'tests/'.

A comprehensive 'requirements.txt' was formulated and installed, providing the foundation for numerical solvers, machine learning, and control components (including NumPy, SciPy, Numba, JAX, and CasADi).

2. Phase 1: Physics Engine Implementation

The core physical models dictating wellbore and reservoir behavior have been successfully coded:

- Reservoir Module: Implemented the Marx-Langenheim model for steam injection heated volume, and the Boberg-Lantz model for thermal decline during the soak and production phases.
- Viscosity Module: Implemented the ASTM D341 Walther double-logarithmic equation to model heavy oil viscosity as a function of temperature.
- Rod String Module: Coded the Gibbs damped wave equation (accelerated using Numba for fast finite-difference solving) and the Float Margin Index (FMI) to predict safe rod speeds.
- Wellbore Module: Established the baseline for Ramey's transient heat transmission.

3. Mathematical Stability Optimization

During testing, a critical numerical instability was identified in the Marx-Langenheim equation. For long-duration CSS cycles, large dimensionless time (t_D) values caused standard IEEE 754 exponential functions to overflow to infinity, crashing the simulation.

This was resolved by replacing the explicit exponential-error function product with SciPy's 'erfcx' (scaled complementary error function), allowing the simulator to process large timesteps natively and without overflow.

4. Phase 2: Synthetic Data Generation

Built the 'SyntheticDataGenerator' module. This engine leverages the implemented physics core to simulate realistic Cyclic Steam Stimulation (CSS) data over typical 80-day cycles. It outputs time-series data including true and measured reservoir temperatures, dynamic viscosity changes, and declining production rates, augmented with simulated sensor noise to train the learning layers.

5. Verification & Testing

A robust test suite ('tests/test_physics.py') was executed against all Phase 1 physics modules. All calculations evaluated successfully, boundaries held within expected physical limits, and the environment

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reported 100% test passage. The system is now ready for Phase 3 (Data Assimilation).